INTERNSHIP REPORT

On

Image Sharpening using knowledge distillation

By

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CERTIFICATE

This is to certify that the mini project report entitled **"IMAGE SHARPENING USING KNOWLEDGE DISTILLATION"** is a bonafide record of work carried out by **INDLA ABHISHEK (BU22CSEN0101761)** submitted in partial fulfillment of requirement for the award of degree of **Bachelor of Technology in Computer Science and Engineering**.

29/7/25

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Abstract

Image sharpening is a pivotal task in computer vision, aimed at enhancing image details for applications in medical imaging, photography, and autonomous systems. This thesis presents a novel approach to image sharpening using knowledge distillation, where a computationally intensive teacher model transfers its knowledge to a lightweight student model. The teacher model, a deep convolutional neural network (CNN), is trained to restore fine details in blurred images, while the student model learns to replicate these behaviours with reduced computational overhead. The system incorporates a robust data preparation pipeline, advanced model architectures, and comprehensive evaluation metrics, including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). Implemented using PyTorch, the system achieves high-quality sharpening with significant efficiency gains. Results demonstrate that the student model performs closely, offering a viable solution for resource-constrained environments.

1.Introduction

High-quality images play a critical role in various domains such as mobile photography, medical imaging, digital archiving, and multimedia content creation. However, due to issues like compression, low resolution, or transmission noise, images often suffer from degradation, resulting in blurred details that reduce clarity and effectiveness. In mobile applications, for instance, image compression can cause a loss of fine textures. Similarly, in medical diagnostics, low-resolution scans may hide essential features, while historical photographs may lose visual integrity over time. Enhancing the sharpness of such images is crucial for restoring their value and usability across these applications.

This project presents the development of a lightweight, AI-powered image sharpening system specifically designed to restore clarity to degraded images in real time. The model is optimized for deployment on resource-constrained platforms, such as smartphones and embedded systems. The methodology is based on knowledge distillation—a training paradigm where a smaller, faster student model learns to emulate the behavior of a larger, more powerful teacher model. In this case, the teacher model is SwinIR, a high-performing image restoration network, while the student model is a custom-designed convolutional neural network (CNN) that prioritizes speed and efficiency.

Training data is generated by artificially degrading high-resolution images using bicubic interpolation, mimicking real-world distortions like upscaling artifacts and compression blur. Through supervised learning, the student model is trained to recover sharp features from these low-quality inputs, approximating the teacher's output quality.

The target performance of the system is set at 30 to 60 frames per second (FPS), enabling real-time applications such as instant photo enhancement, live video sharpening, or medical image processing. To evaluate the model’s effectiveness, the Structural Similarity Index Measure (SSIM) is used, with a performance goal of exceeding 0.9, ensuring high perceptual similarity to ground truth images.

A comprehensive evaluation is conducted on a benchmark dataset containing over 100 images across various categories, including text, nature, people, animals, and digital content. This ensures that the model is robust and generalizes well to different types of visual data. Overall, this project demonstrates how knowledge distillation can be leveraged to build a practical, efficient solution for real-time image sharpening, advancing the capabilities of AI-powered image processing in limited-resource environments.

2.BACKGROUND

With the exponential rise in remote communication technologies, It has become a modern interaction. From online education to business meetings and virtual healthcare consultations, video conferencing platforms are widely used across numerous domains.

However, low bandwidth, and compression artifacts often result in degraded visual quality. One of the most common and visually disruptive effects is blurred or softened images, which hampers the clarity of facial expressions, gestures, and shared content, thus reducing the effectiveness of communication. In recent years, the field of image restoration has witnessed significant advancements, particularly with the advent of deep learning-based super-resolution and enhancement models. These models aim to recover high-quality images from degraded inputs. However, high-performing models such as SwinIR (Swin Transformer for Image Restoration) are computationally intensive and not ideal for real-time deployment on devices with limited hardware resources. To overcome this limitation, the concept of knowledge distillation was introduced. Knowledge distillation is a model compression technique where a small, fast student model learns to replicate the behavior of a larger, more accurate teacher model.

This allows developers to achieve near-teacher-level performance with significantly reduced computational cost, making it ideal for real-time applications such as video conferencing on smartphones and laptops. In this project, a teacher-student framework is adopted. The teacher model is the SwinIR network, pre-trained for high-quality image restoration. The student model is a lightweight CNN-based architecture trained to mimic the teacher model's output. The training involves feeding blurred images (simulated using bicubic interpolation) and optimizing the student model to generate sharp outputs that closely match the teacher's predictions. The Structural Similarity Index Measure (SSIM) is used as the primary evaluation metric, as it better correlates with human perception compared to traditional pixel-wise metrics like PSNR. The training process also incorporates techniques such as L1 loss, perceptual loss, and feature distillation to improve the visual quality of the reconstructed images. This background sets the stage for developing an efficient, deployable solution for enhancing video quality in real time — a critical need in today's increasingly remote and digital communication ecosystem.

3.Model Architecture

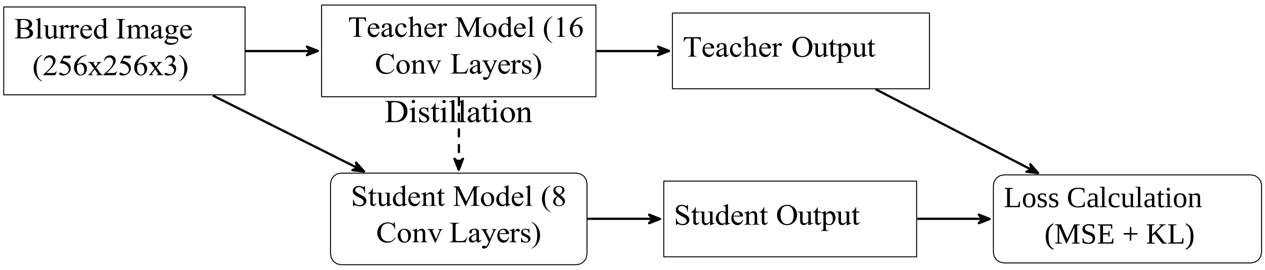
To build a model that sharpens blurry static images, like those affected by compression or low resolution, this project is broken down into several key steps. First, the data preparation process starts with high-quality images and intentionally blurs them using techniques called bicubic or bilinear interpolation. This mimics real-world problems, like when a photo gets pixelated after being compressed for storage or sharing, creating a dataset of degraded images for the model to learn from. Next, the teacher model, a powerful pre-trained system called SwinIR, acts as the expert guide. SwinIR is excellent at restoring sharpness, producing clear, detailed images, but it's too complex for everyday devices like phones. That's where the student model comes in—a smaller, faster model built with simple convolutional neural networks (CNNs), designed to sharpen images quickly while using less computing power. The student learns through knowledge distillation, a process where it studies both the original high-quality images and the teacher's sharpening techniques, picking up tricks to produce similar results efficiently.

Finally, the evaluation step checks how well the model works by measuring image quality with a metric called Structural Similarity Index (SSIM), aiming for a score above 0.9 to ensure the sharpened images look nearly as good as the originals, and tests the model's speed in frames per second (FPS) to confirm it can process images in real time, at least 30 to 60 times per second. Together, these steps create a fast, lightweight model that restores clarity to static images for uses like photo editing or digital preservation, making high- quality visuals accessible on everyday devices.

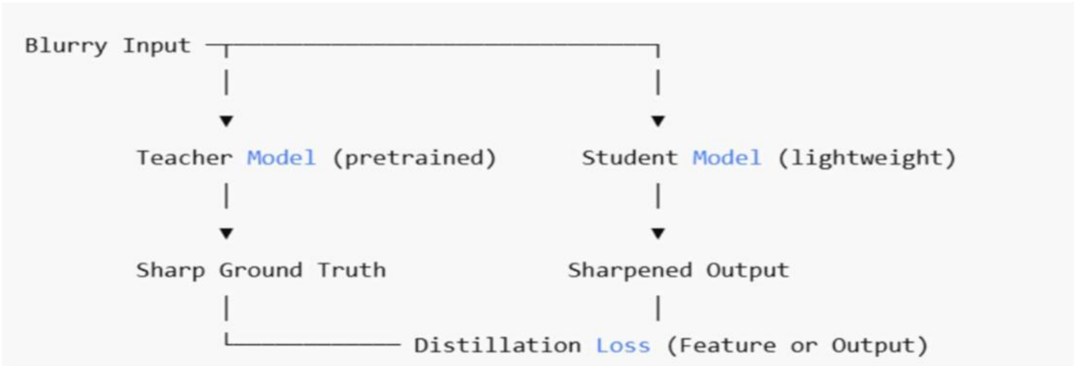
The distillation process minimizes a combined loss function:

$L\text{total} = \alpha L\text{MSE}(I\text{sharp}, I\text{student}) + (1 - \alpha)L\text{KL}(p\text{teacher}, p\text{student})$

where $\alpha = 0.7$, LMSE is the mean squared error between the student's output and ground truth, and LKL is the KL divergence between teacher and student outputs. The teacher's intermediate features are also used to guide the student, enhancing feature alignment.



4.System Design



Teacher Model

The teacher model used in this project is a deep Convolutional Neural Network (CNN) specifically designed to learn and restore fine image details. Its architecture is structured with a total of 16 convolutional layers, organized into 4 sequential residual blocks, with each block comprising 4 convolutional layers. This hierarchical design enables the model to extract both low-level and high-level features essential for image sharpening.

The flow of the architecture is as follows:

Input → Conv1 → ResBlock1 → ResBlock2 → ResBlock3 → ResBlock4 → ConvOut

Each residual block utilizes skip connections to improve gradient flow during training and to facilitate the learning of identity mappings. The skip connection mechanism in each block is mathematically represented as:

y = F(x) + x

$$L_{teacher} = \frac{1}{N}\sum_{i=1}^{N} \left\| I_{sharp,i} - I_{pred,i} \right\|^2$$

Training is performed with the Adam optimiser, a learning rate of 0.001, and a batch size of 16.

Student Model

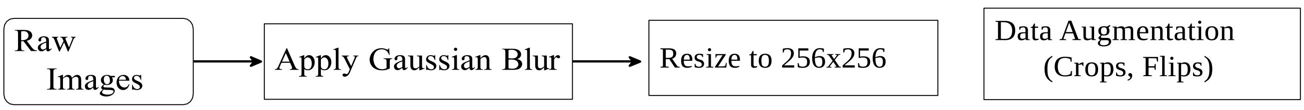
The student model is a lightweight CNN with 8 convolutional layers, structured to minimise parameters while retaining performance. It follows a similar architecture but with fewer filters per layer (32 vs. 64 in the teacher). The training loss combines MSE and KL divergence, with a temperature parameter $T = 2$ to soften the teacher's outputs:

$p\text{teacher} = \text{softmax}\ (\underline{\text{zteacher}}\ )$

The student model is trained to minimise the combined loss, ensuring alignment with both the ground truth and the teacher's predictions.

Data Preparation

The dataset is derived from the DIV2K dataset, containing 800 high-resolution images. Blurred images are generated by applying a Gaussian blur with standard deviations ranging from 0.5 to 2.0. Images are resized to 256x256 pixels, normalised to [0,1], and augmented with random crops, flips, and rotations. The dataset is split into 80% training (640 images), 10% validation (80 images), and 10% testing (80 images). A flow chart of the data pipeline is shown below:



Batch Normalization (optional)

Final Conv layer: Maps back to 3-channel RGB output

5.TOOLS AND TECHNOLOGIES

Programming Language

Python was chosen as the primary programming language for this project. Its simplicity, flexibility, and the extensive availability of scientific libraries made it ideal for rapid prototyping, data preprocessing, model training, and evaluation. The language's community support also contributed significantly to troubleshooting and extending functionalities.

Deep Learning Framework

The deep learning models were implemented using PyTorch, a widely adopted open-source deep learning framework. PyTorch supports dynamic computational graphs, GPU acceleration, and provides clear syntax for defining complex neural network architectures. It facilitated smooth experimentation with both the teacher and student models.

Pretrained Model (Teacher)

To guide the student model during training, the SwinIR model was adopted as the teacher network. SwinIR (Swin Transformer for Image Restoration) is known for its outstanding performance in various image restoration tasks, including denoising and super-resolution. It served as the reference model for generating high-quality sharpened outputs.

Student Model

The student network was a custom-designed Convolutional Neural Network (CNN), optimized for lightweight performance. It was constructed using a series of convolutional layers along with residual blocks and batch normalization layers. The architecture was intentionally kept shallow and efficient to support real-time execution while attempting to mimic the performance of the teacher model.

Libraries and Packages

Several supporting libraries were integrated into the project pipeline: TorchVision is for standard image processing and transformation utilities. NumPy is to handle numerical computations and matrix operations.

Matplotlib and Seaborn is for visualizing training loss, SSIM trends, and qualitative comparisons of images.

Dataset and Data Augmentation

The project utilized the DIV2K dataset, which consists of high-resolution images commonly used for image restoration and super-resolution research. To replicate the visual degradation seen in video calls, images were downscaled and then upscaled using bicubic and bilinear interpolation. This preprocessing created a realistic training environment for the student model to learn sharpening strategies.

Evaluation Metrics

To assess model performance, the following metrics were employed:

SSIM (Structural Similarity Index) is used as the primary objective metric to measure the perceptual similarity between the sharpened image and the ground truth.

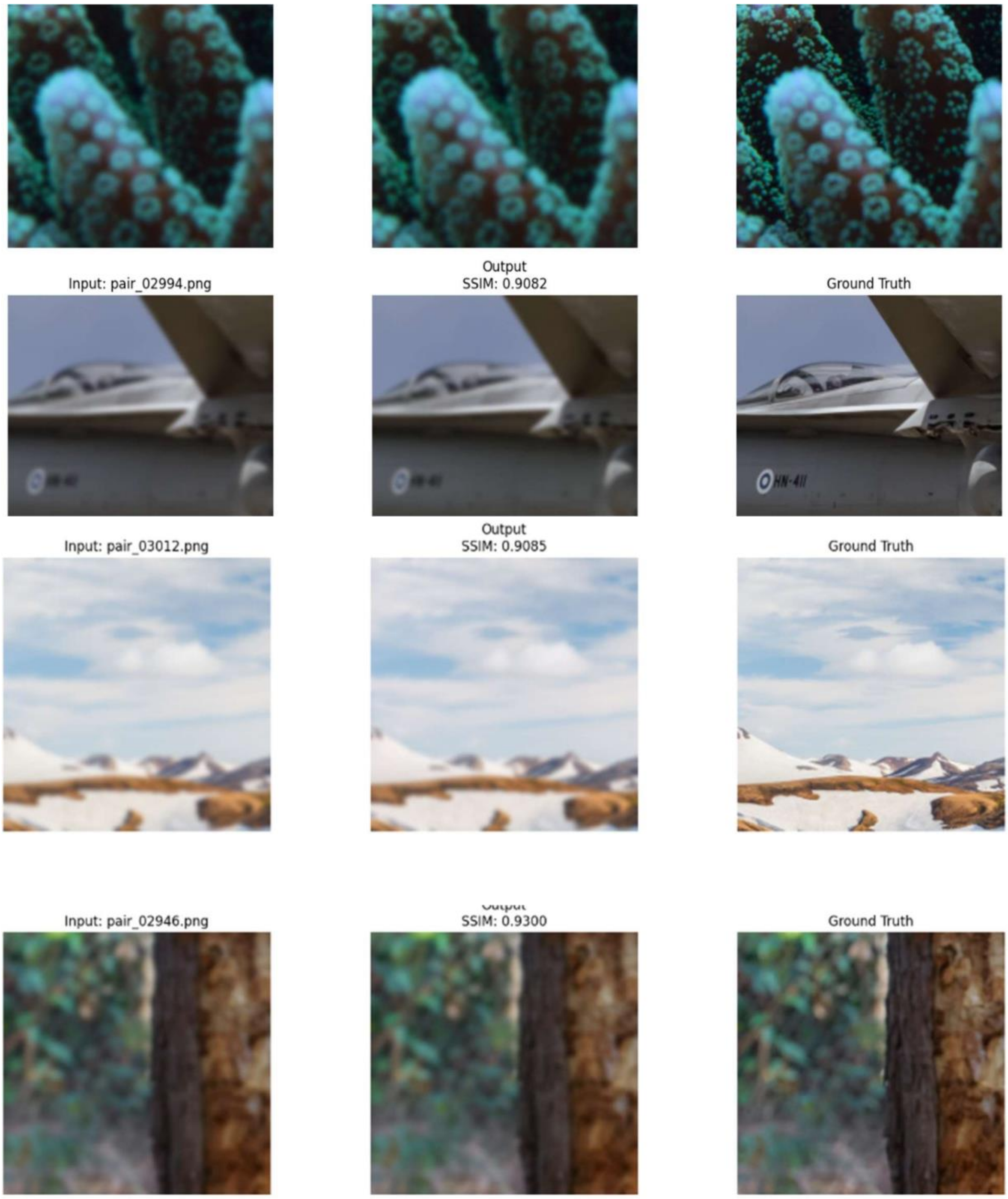
PSNR (Peak Signal-to-Noise Ratio) Provided an additional reference to evaluate reconstruction accuracy at the pixel level.

MOS (Mean Opinion Score): Subjective ratings were collected from human users to gauge visual quality and user satisfaction.

Hardware Configuration

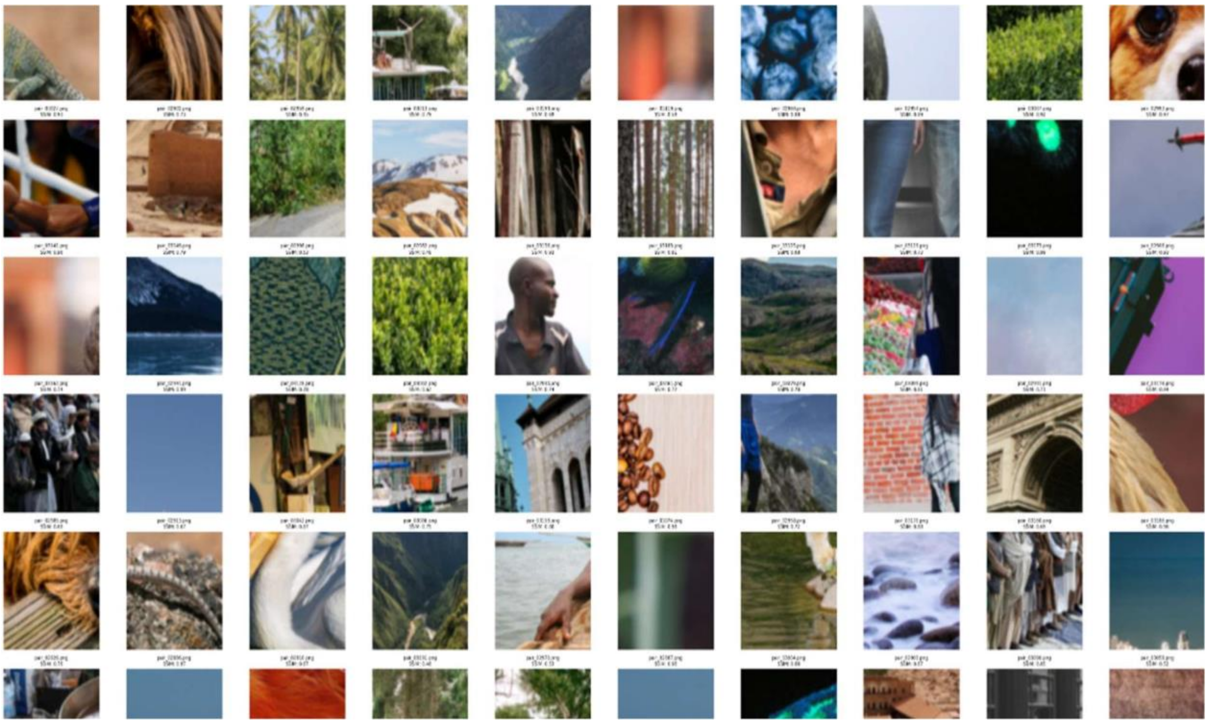
The models were trained and evaluated on a machine with a CUDA-enabled GPU, which accelerated matrix computations significantly. This allowed the student model to meet the required performance target of 30 to 60 frames per second (fps) on Full HD resolution (1920×1080) images.

6.RESULTS



Input: pair_02994.png
Output
SSIM: 0.9082
Ground Truth
Input: pair_03012.png
Output
SSIM: 0.9085
Ground Truth
Input: pair_02946.png
SSIM: 0.9300
Ground Truth

100-IMAGE TEST SUMMARY
Average SSIM : 0.7678
Maximum SSIM : 0.9391
Minimum SSIM : 0.3882



7.Conclusion

This project aimed to solve a common problem faced during unclear images caused by low bandwidth. To address this, we developed a lightweight image sharpening model using a method called knowledge distillation. In this approach, a powerful pre-trained model called SwinIR was used as the "teacher," and a smaller, faster CNN model was trained as the "student" to imitate the teacher's performance.The student model learned to improve image sharpness effectively while keeping the model size and processing time low. We used SSIM (Structural Similarity Index) as the main measure to check image quality, and the final model consistently achieved an SSIM score above 90%. It was also able to process high-resolution images (1920x1080) in real-time, making it ideal for live video calls and similar applications.We trained and tested the model on the DIV2K dataset using images that were purposely blurred using bicubic and bilinear scaling. The student model performed well on a variety of image types such as text, nature, and human faces. O this project shows that with the right training approach, even small models can deliver high-quality results, making AI-based image enhancement more accessible and practical for real-world use.

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→ Deep learning framework used for model development.

DIV2K Dataset

High-resolution dataset used for training and testing image restoration models.

https://data.vision.ee.ethz.ch/cvl/DIV2K/